$\boxed{\text{Solution:}}$ p: n^3+5 is an odd integer q: n is an even integer.

a) We will show that $p \rightarrow q$ is True by contraposition.

$\therefore$ We will prove $\neg q \rightarrow \neg p$ is True.

$\neg p$: n^3+5 is not odd, i.e. n^3+5 is even (Defn. 1)

$\neg q$: n is not even, i.e. n is odd (Defn. 1)

Let $\neg q$ be True. $\boxed{\therefore n \text{ is an odd integer}}$. By Defn 1, $\exists$ integer K such that $n = 2K+1$. $\therefore n^3+5 = (2K+1)^3+5 = 8K^3+12K^2+6K+6 = 2(4K^3+6K^2+3K+3)$

$\because K$ is an integer, $\therefore (4K^3+6K^2+3K+3)$ is an integer.

By Defn 1, n^3+5 is an even integer. $\therefore \neg p$ is True.

By Defn. of conditional, $\neg q \rightarrow \neg p$ is True. $\therefore p \rightarrow q$ is True.

b) We will show that $p \rightarrow q$ is True by contradiction.

$\therefore$ We will prove $(p \land \neg q) \rightarrow F$ is True.

Let $(p \land \neg q)$ be True. By defn. of AND, p is True, $\neg q$ is True.

From part (a), we can say that both n^3+5 and n are odd integers

By defn 1, $\exists$ integers K_1 and K_2 such that $n^3+5 = 2K_1+1$ and $n = 2K_2+1$

$\therefore (n^3+5) - n = (2K_1+1) - (2K_2+1)$ or, $(n^3+5) - n = 2K_1 - 2K_2$

or, $(n^3+5) - n = 2(K_1 - K_2)$. It's easy to see that (K_1-K_2) is an integer.

$\therefore \boxed{(n^3+5)-n}$ is an even integer, by Defn. 1

$(n^3+5) - n = (n^3-n)+5 = n(n^2-1)+5 = n(n-1)(n+1)+5.$

Substituting $n = 2K_2+1$, $(2K_2+1)(2K_2)(2K_2+2)+5$

$= 2((2K_2+1)(K_2)(2K_2+2)+2)+1$